Changes in Supermarket Access for Seniors and SNAP Households: An Exploratory Data Analysis of Urban and Rural Census Tracts, 2015-2019

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Background and Data

1.1 Data set and Source of Data

Our group used the 2015 and 2019 Food Access Research Atlas data sets, available through the U.S. Department of Agriculture Economic Research Service.

1.2 Why does this data set matter?

The data sets are of interest because they provide insight into access to supermarkets and essential food resources across different communities in the United States of America. They can be used to examine food access among people receiving benefits, those living in areas of high deprivation or low socioeconomic status, those living further away in urban vs rural areas, and different demographic groups. Using data from multiple years also allows us to identify changes over time and determine whether there have been noticeable improvements in food access.

1.3 Types of Data in the data set

The data sets contain information including census tract, poverty rate, food access at specified distances, and access measures across different racial and age groups, as well as SNAP beneficiaries. Each observation represents a U.S. census tract, with columns containing demographic, socioeconomic and food-access measures associated with that tract. The majority of variables are numerical, including population counts, poverty rates and food-access measures, while categorical variables include state and county. The data sets are cross-sectional snapshots for 2015 and 2019 rather than continuous time-series data. Individual variables represent specific measures of food access for particular demographic groups and distance thresholds.

1.4 Completeness of Data set

One point of interest was the substantial amount of missing data in the 2019 data set. A number of observations were originally represented as NULL. Although some of these appeared as though they could represent zero values, there was insufficient information to confidently distinguish a true zero from a missing observation. To avoid introducing unsupported assumptions, all NULL and other missing values were therefore converted to NA. This provided a consistent representation of missingness and allowed R to appropriately exclude these observations where necessary during subsequent analysis.

Table 1: Number of missing Senior and SNAP low-access share observations, by year.

Year	Senior Missing	SNAP Missing
2015	0	0
2019	29953	29918

Missingness was not evenly distributed between the data sets. For the two variables central to our analysis, the 2015 data set contained no missing observations, while the 2019 data set contained 29,953 missing observations for the senior low-access percentage and 29,918 for the SNAP low-access percentage. This substantial difference in completeness between years was therefore considered when interpreting comparisons between 2015 and 2019. Comparisons between the data sets were done with caution.

Table 2: Missing Senior and SNAP low-access observations by urban and rural census tracts in 2019.

Area	Census Tracts	Senior Missing	Senior Missing (%)	SNAP Missing	SNAP Missing (%)
Rural	17362	9970	57.4	9958	57.4
Urban	55169	19983	36.2	19960	36.2

Further inspection identified a clear relationship between missingness and urban/rural classification in the 2019 dataset. Although urban census tracts contained a greater number of missing observations overall, this reflects the much larger number of urban tracts in the dataset. Proportionally, 57.4% of rural census tracts had missing values for both the Senior and SNAP low-access measures, compared with 36.2% of urban census tracts. This indicates that missingness is disproportionately concentrated among rural observations and should be considered when interpreting subsequent urban-rural comparisons.

Similarly, both data sets also contained a considerable number of genuine NA values, particularly for SNAP_la_rate (2,405) and senior_la_rate (520).

1.5 Integration of data set

To enable comparisons between 2015 and 2019, both data sets were cleaned and standardised before integration. Although the data sets had largely consistent structures, variable names were first converted to lowercase to account for minor differences in capitalisation between years. Equivalent variables relevant to our analysis were then selected from each data set and converted to consistent data types. Missing and NULL values were also represented consistently as NA.

Further transformations were required to make the food-access measures directly comparable. An Urban/Rural classification and year indicator were created, while Senior and SNAP low-access measures were constructed using the relevant distance thresholds for urban and rural census tracts. Percentage measures also required standardisation because the 2015 share variables were represented as proportions, while the corresponding 2019 variables were already expressed as percentages. Once the variables were consistently defined and formatted, the data sets could be vertically combined for comparison between years.

Table 3: Overview of the 2015 and 2019 Food Access Research Atlas datasets.

Feature	2015	2019
Census tracts	72864	72531
Variables retained for analysis	25	25
Urban tracts	55172	55169
Rural tracts	17692	17362
Senior access measures	Yes	Yes
SNAP access measures	Yes	Yes
Poverty measures	Yes	Yes

Ethics, Privacy & Security

2.1 Ethics

This project involves data describing vulnerable populations, specifically seniors aged 65 and over and SNAP-receiving households. These populations require discretion as careless findings or framing about them can further stigmatise affected communities. A key consideration for this analysis is avoiding deficit oriented framing, practically, this means describing a tract as "low access" reflects a structural and geographic outcome specifically the distance for a household to supermarkets or transport infrastructure. It is not intended to signify the choices of the individuals living in the household, this is to avoid perpetuating any stereotypes or stigmas. As a result the language of the report has been mindfully selected to reflect this. For example we would state "A tract has a high rate of low-access seniors" instead of language that could be read as describing a senior or snap receiving households like "seniors in this area don't shop at supermarkets"

Another ethical risk we are considering is an "ecological fallacy", which is the error of drawing conclusion about individuals based on patterns observed in aggregated group level data (Robinson, 1950). All analysis in this project is conducted at the census-tract level which uses aggregated rates for an area rather than individual data. So where a tract is classified as "low access" indicates the aggregate population meets USDA distance thresholds, it does not mean that every individual within that tract lacks access. As a result our conclusions are based on patterns observed in tracts, not on individuals. Additionally, A further ethical consideration of this relates to the integrity of comparing census tracts between the 2015 and 2019 datasets. Changes between the data sets could show a real change in food access but could also reflect changes in census boundaries or the collection methodology between years. As a result of this ethical consideration we treat the differences from 2015 to 2019 as descriptive patterns not exactly a causal change, this limitation is further discussed in section 3.4.

2.2 Privacy

This project uses only publicly available, government-released data (USDA Food Access Research Atlas and U.S. Census Bureau data), both provided at the census-tract level. We did no original data collection of our own, or have any interaction with the individuals involved with the data. The data is already aggregated tract-areas and released for public use by the U.S. government, so the privacy risk to individuals is minimal. The risks to privacy is not technical like how the data is structured or analysed but in the interpretation of the data. There is still a risk (although very small risk) of individual identification for tracts with very small populations e.g a small amount of senior SNAP households in a rural tract, where in principle it could be easier to infer information about specific households. This represents a limitation of the dataset from a privacy perspective, though it is a low and indirect risk rather than a risk introduced by our analysis.

This residual privacy risk is relevant to the pattern we identified in section 1.4 about missingness in the 2019 dataset which was disproportionately concentrated among rural census tracts: 57.4% missing compared with 36.2% in urban tracts, as shown in table 2. This pattern suggests the data is not missing completely at random, missingness appears systematically related to tract characteristics such as urban/rural status (Rubin, 1976). One plausible explanation is disclosure-avoidance suppression, as rural tracts tend to have smaller and more sparsely populated populations, meaning aggregated counts may be more susceptible to suppression to protect individual privacy. Disclosure avoidance is a strategy commonly used to protect anonymity in low-population tracts. While we cannot directly confirm that this specific mechanism caused the missingness pattern observed here, it represents one plausible, consistent explanation, rather than a flaw introduced by our own data processing.

2.3 Security

As mentioned in the privacy section, this project uses only public, pre-aggregated data, so the overall security risk associated with this project is low. We practiced secure data management by storing all working data and code within a shared, private GitHub repository, accessible only to the members of our group, rather than a public repository or any other external sharing method. This gave us a controlled, access-limited location for our data, code, and analysis, alongside full version control over changes made throughout the project

Beyond access control, this repository also supports reproducibility. Every data cleaning, transformation, and analysis step used in this project, like reading in the raw datasets, standardising variable names, constructing the urban/rural classification, and matching census tracts between 2015 and 2019 is all recorded within the rmd file. This means that the entire analysis can be reproduced end-to-end by any member of the group or authorised person simply by re-running the document. This makes results traceable and easily backed up with evidence provided by the source itself.

With some additional time and resources to allocate into security it would be worthwhile establishing a documented data deletion plan for after the project concludes.

Exploratory Data Analysis

3.1 Overview of the Analysis

The exploratory analysis focused on the low-supermarket-access shares for two groups: adults aged 65 years and over, and SNAP-receiving households. Comparisons were made between 2015 and 2019 and between urban and rural census tracts. For urban census tracts, low supermarket access was measured using the one-mile threshold, while the ten-mile threshold was used for rural census tracts. All low-access shares were expressed as percentages to ensure consistency between the 2015 and 2019 datasets.Because urban and rural low-accessibility percentages are not calculated on the same scale, the report's urban-rural comparisons reflect each area's results under USDA standards—not a direct distance comparison. This analysis covers 2015–2019 changes and urban-rural tract differences, with all figures in percentages for consistent year-over-year comparison.

Table 4 shows a clear urban-rural gap: the mean low-access share for older adults in urban census tracts is roughly double that in rural tracts (3.4% vs. 1.7% in 2015; 5.3% vs. 4.0% in 2019), and the same pattern holds for SNAP households (1.3% vs. 2.9% in 2015; 2.8% vs. 4.3% in 2019). This pattern reflects the differing USDA thresholds applied to urban and rural areas, rather than an absolute access gap. The effective sample size dropped substantially between 2015 and 2019, with rural tract observations more than halving (from 17,692 to around 7,400), suggesting that 2019 estimates should be interpreted with greater caution than those for 2015. Based on the table, every mean exceeds its corresponding median, with the largest gap observed in the 2015 rural group (mean 1.7% vs. median 0.0%), indicating a strongly right-skewed distribution where a small number of tracts with particularly severe accessibility barriers pull the averages upward.

Table 4: Mean and median low-access share (%) for seniors and SNAP households, by area and year.

Area	Year	Senior Mean	Senior Median	Senior N (Valid)	SNAP Mean	SNAP Median	SNAP N (Valid)	N Tracts
Rural	2015	1.7	0.0	17692	1.3	0.0	17692	17692
Rural	2019	4.0	1.7	7392	2.8	1.0	7404	17362
Urban	2015	3.4	0.6	55172	2.9	0.3	55172	55172
Urban	2019	5.3	3.5	35186	4.3	1.8	35209	55169

3.2 Senior Low-Access Share: Distribution and Urban–Rural Comparison

The following sections look at distribution of variables in more detail, starting with seniors.

Figure 1 shows the distribution of senior low-access shares by year and urban/rural areas. The distributions were strongly right-skewed across all four groups, with most census tracts having relatively low senior low-access shares and a smaller number having much higher values. Urban census tracts generally showed higher low-access shares than rural census tracts. The 2019 distributions also appeared shifted toward higher values than in 2015. However, interpret the 2019 results cautiously because the senior low-access variable contained substantial missing data.

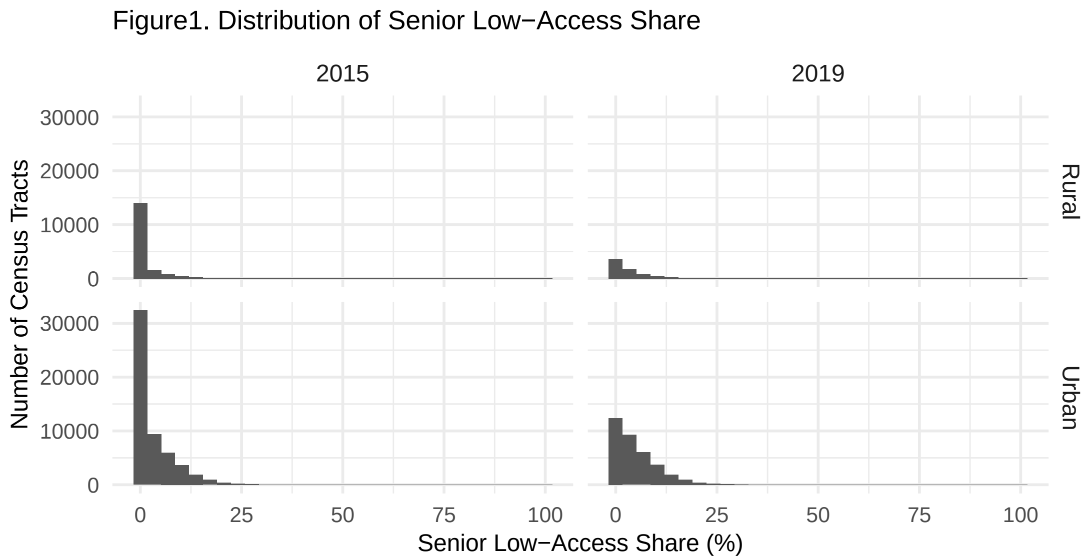
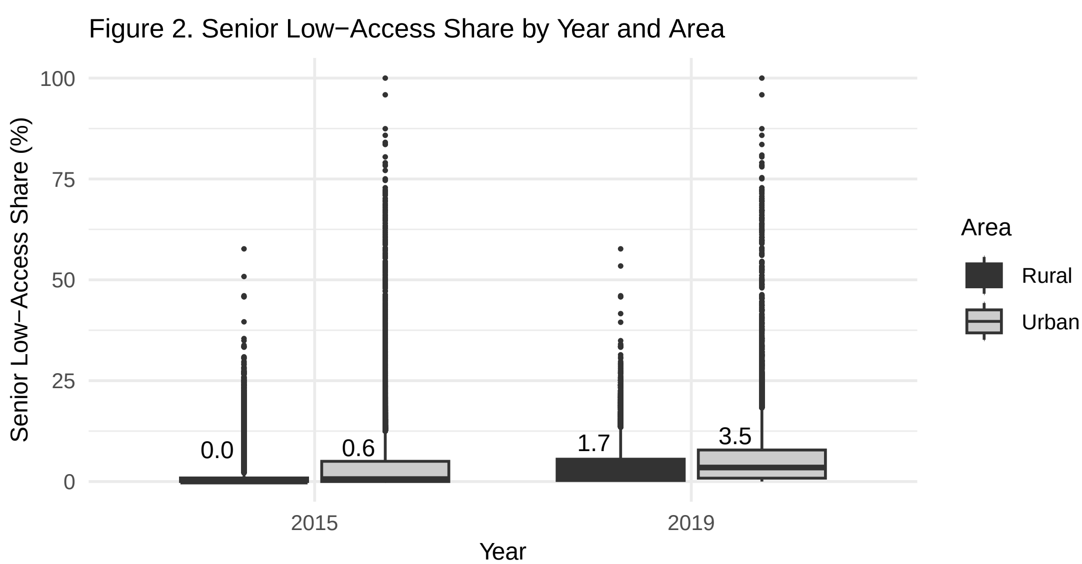


Figure 2 provides a more direct comparison between urban and rural tracts for the senior low-access share. In both 2015 and 2019, the median senior low-access share was consistently higher in urban tracts than in rural tracts, rising from 0.6% to 3.5% in urban areas compared with 0.0% to 1.7% in rural areas. The spread also widened between the two years: in rural areas, the boxp shifted from being almost flat at zero in 2015, but by 2019 it had extended noticeably upward by several percentage points. Urban tracts showed a similar pattern, with the box stretching further above the median, indicating greater variability across tracts in 2019. In both years, a number of outlying high values are visible, which is consistent with the right-skewed distribution observed in Figure 1.



Overall, the senior low-access share was generally higher in urban than rural census tracts, and the distributions were strongly right-skewed in both years. The apparent increase from 2015 to 2019 may be partly attributable to differences in the set of census tracts with available data, rather than reflecting a genuine change. In 2019, values for low accessibility among older adults were missing for approximately 30,000 census tracts. This issue is addressed directly in the matched analysis presented in Section 3.4.

3.3 SNAP Low-Access Share: Distribution and Urban–Rural Comparison

This section examines the SNAP low-access share in more detail.

Figure 3 displays the distribution of SNAP low-access share by year and urban-rural regions. The distributions of all four groups exhibit a strong right-skewed pattern: the SNAP low-access share in most census areas are relatively low, while only a few census areas have significantly higher values. Urban census areas show a longer and more extended tail to the right in both years compared to rural census areas, indicating greater variability among urban census areas.

Figure 3. Distribution of SNAP Low−Access Share

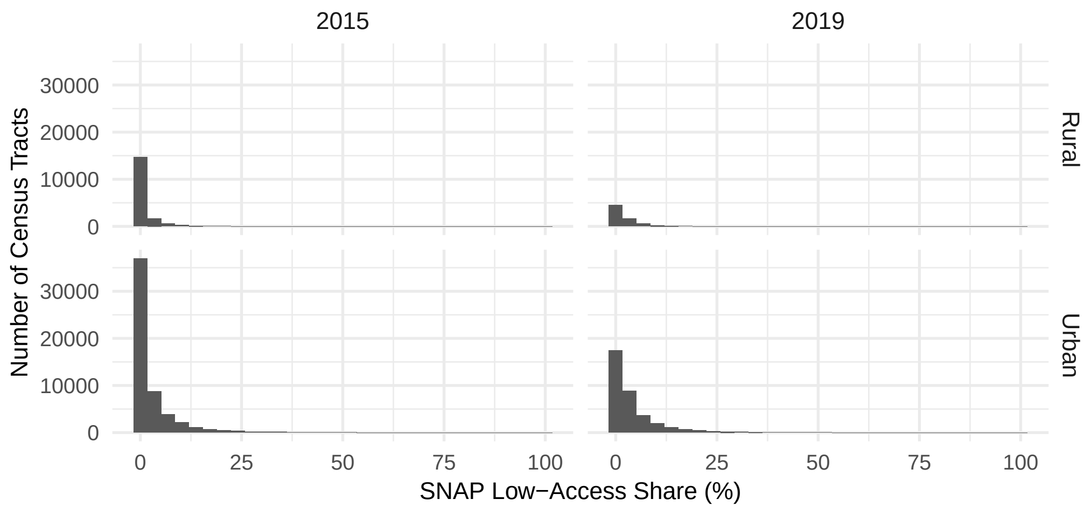
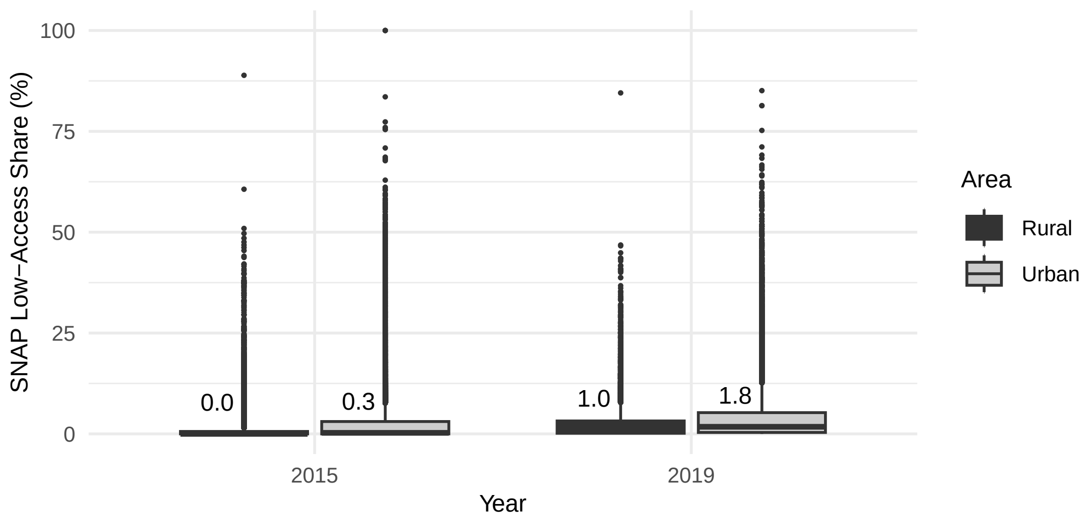


Figure 4 shows that the median SNAP low-access share was consistently higher in urban tracts than in rural tracts in both years, rising from 0.3% to 1.8% in urban areas compared with 0.0% to 1.0% in rural areas. The spread also widened in both areas: rural boxes expanded from nearly flush with zero to about one percentage point higher, while urban boxes stretched further above the median, indicating greater tract-level variation in 2019. The distribution in both years and for both regions remained strongly skewed to the right, with high values observed extending to approximately 85% - 100%, indicating that a small portion of the surveyed areas were more pronounced in urban areas and continued to face significantly higher barriers to supermarket accessibility compared to typical surveyed areas.

Figure 4. SNAP Low−Access Share by Year and Area



In short, the SNAP low-access share was generally higher in urban than rural census tracts, and the distributions were strongly right-skewed in both years. The apparent increase from 2015 to 2019 may be partly attributable to differences in the set of census tracts with available data, rather than reflecting a genuine change. In 2019, values for SNAP low-access were missing for approximately 29,918 census tracts. This issue is addressed directly in the matched analysis presented in Section 3.4.

3.4 Changes between 2015 and 2019

To examine changes between 2015 and 2019 more directly, census tracts in the two datasets were matched using their Census Tract identifiers. A total of 72,506 census tracts were successfully matched across the two years. Of these, 21 census tracts changed their urban/rural classification between 2015 and 2019. These tracts were excluded from the change analysis because the low-access distance threshold differs between urban and rural areas. This resulted in 72,485 census tracts with a stable urban/rural classification.

For the paired analysis, only census tracts with non-missing low-access shares in both 2015 and 2019 were included. Of the 72,485 tracts with stable classification, 42,554 had complete senior data in both years and 42,589 had complete SNAP data in both years (Table 5). Change was calculated as: `Low-access share2019 - Low-access share2015` and is reported in percentage points. Positive values therefore represent an increase in low-access share between 2015 and 2019, while negative values represent a decrease.

Table 5 reveals an interesting contrast: the trends for senior and SNAP households moved in opposite directions between 2015 and 2019. The mean proportion of older adults rose slightly in both rural (from 3.94% to 3.96%) and urban (from 5.21% to 5.32%) census tracts, whilst the mean proportion of SNAP recipients fell slightly in both rural (from 2.91% to 2.77%) and urban (from 4.40% to 4.33%) census tracts. These opposite trends suggest that changes in supermarket accessibility for these two groups are not driven by the same underlying trend, indicating that the factors influencing changes in accessibility for older people may not be entirely the same as those affecting SNAP households. However, the magnitude of these changes is small relative to the degree of variation within each group shown in Table 4, and the median changes for all four groups are exactly zero.

Table 5: Paired summary of Senior and SNAP low-access shares, 2015-2019

Group	Area	N	Mean 2015 (%)	Mean 2019 (%)	Median 2015 (%)	Median 2019 (%)	Mean Change (pp)	Median Change (pp)
SNAP	Rural	7399	2.91	2.77	1.00	0.96	-0.14	0.00
SNAP	Urban	35190	4.40	4.33	1.76	1.77	-0.07	0.00
Senior	Rural	7387	3.94	3.96	1.56	1.72	0.02	0.00
Senior	Urban	35167	5.21	5.32	3.35	3.47	0.11	0.00

Figure 5 shows that changes in senior low-access share were strongly concentrated around zero in both rural and urban census tracts. This is consistent with the zero median changes reported in Table 5. Although most census tracts experienced relatively small changes, a smaller number showed substantially larger positive or negative changes.

Figure 5. Distribution of Changes in Senior Low−Access Share (2015−2019)

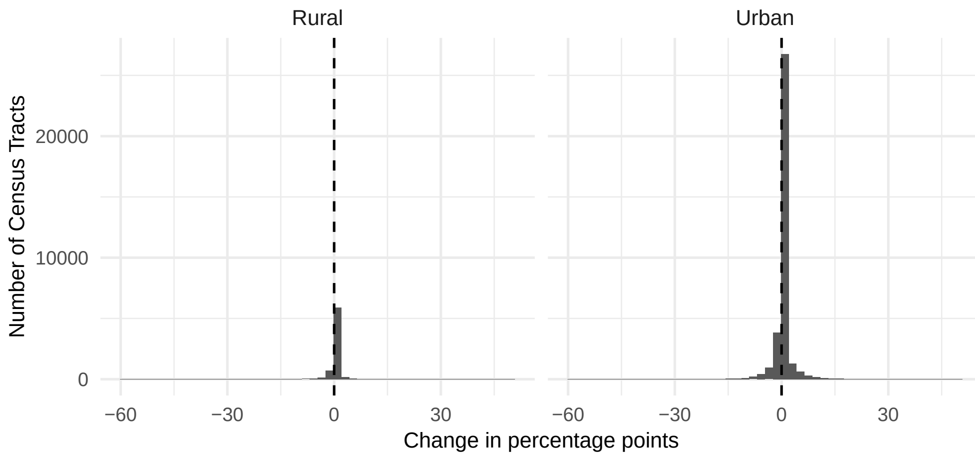
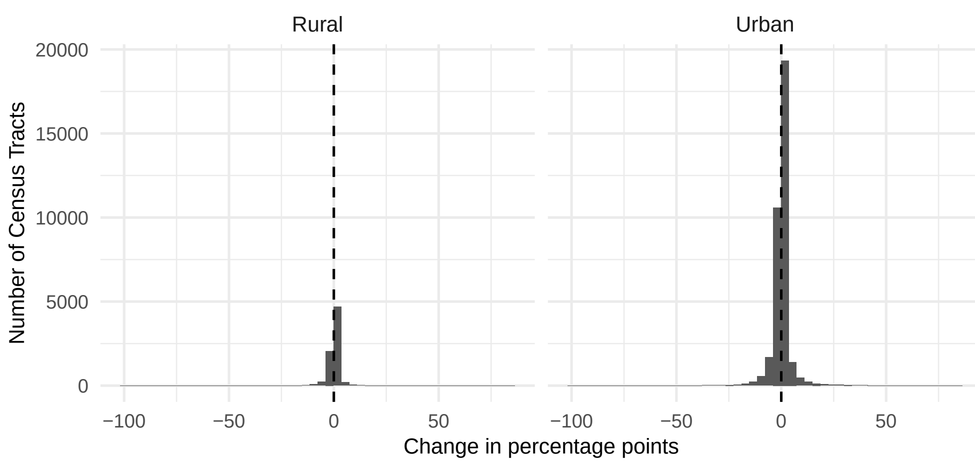


Figure 6 shows that change distributions in SNAP Low-Access share were also strongly concentrated around zero, although some census tracts experienced larger increases or decreases. This is consistent with the zero median changes reported in Table 5, and together they indicate that the overall changes between the two years were small for most matched census tracts.

Figure 6. Distribution of Changes in SNAP Low−Access Share (2015−2019)



In summary, the paired analysis illustrates that the changes at the census tract level between 2015 and 2019 were generally small. Senior and SNAP low-access shares moved in opposite directions on average, but median changes were zero across all four groups, indicating that the small mean shifts were driven by a minority of tracts with larger changes rather than a broad trend. These findings are descriptive results and do not establish statistical significance; the paired analysis itself cannot eliminate the potential bias caused by missing data. If there are systematic differences between the census tracts with missing 2019 data and those with complete data, the above paired results may still be affected.

3.5 Additional Comparison: Mean Low-Access Share by Group

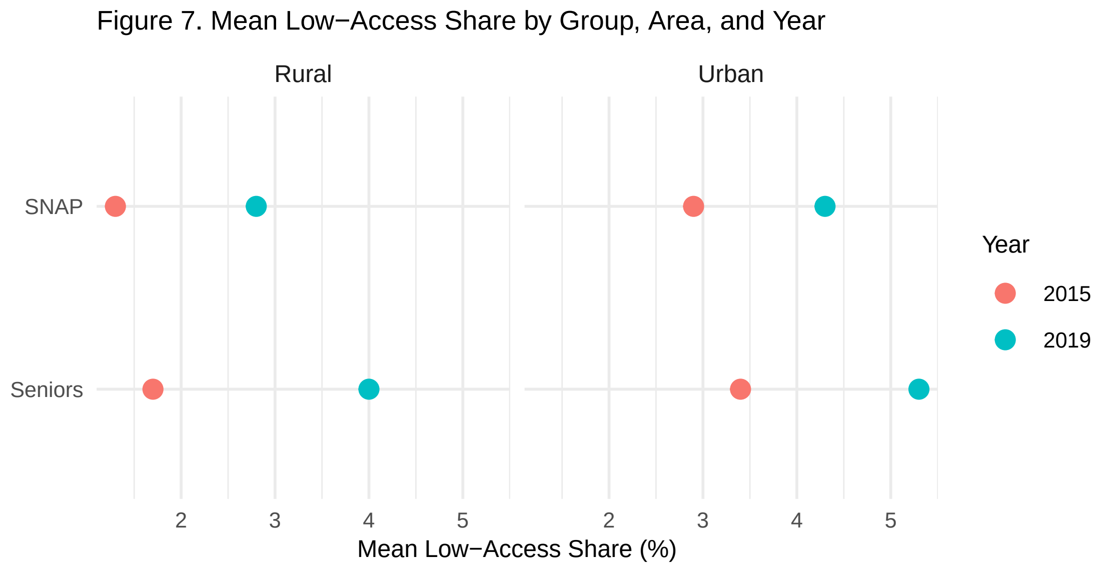


Figure 7 shows that across both Senior and SNAP low-access shares, 2019 shows a higher mean rate than 2015, and Urban tracts show a higher mean rate than Rural tracts, in both years and for both groups

Because "low-access" is defined using a much stricter distance threshold in urban areas (1 mile) than rural areas (10 miles), a higher urban low-access rate does not necessarily mean urban residents travel further to reach a supermarket than rural residents. This may just reflect that urban areas are held to a stricter access standard. A rural resident traveling several miles to a store may still count as "having access" under the rural threshold, while a much shorter distance in a city would count as "low access."

As a result, the lower mean low-access share in rural areas should not be read as evidence that rural food access is genuinely better. Rural residents may need to travel considerably farther in absolute terms before being classified as low-access at all, meaning the rural rate likely understates the true burden of distance compared to the urban rate. This distinction should be kept in mind whenever urban and rural low-access rates are compared directly in this report, rather than treating them as measuring the same underlying concept.

3.6 Summary

In conclusion, the exploratory analysis has identified several important patterns in the indicators of senior and SNAP low-access. First, both low-access share distributions were strongly right-skewed. Most census tracts had relatively low low-access shares, while a smaller number of tracts had substantially higher values. Thus, the mean was generally higher than the median, indicating that the mean was influenced by census tracts with relatively large low-access shares.

Across both measures, urban census tracts consistently showed higher low-access shares than rural tracts in both years. However, since the urban census tracts use a stricter one-mile threshold while the rural census tracts use a ten-mile threshold, this disparity reflects the classification results of each region based on their own USDA standards, rather than directly measuring which group of residents is actually farther from the supermarket.

Although cross-sectional distributions in 2019 appeared to be higher than those in 2015, such comparisons should be interpreted with caution as there were a large number of missing values in the 2019 data.The paired analysis provided a more direct comparison of the same census tracts across years and showed that changes were generally small. Senior low-access shares increased slightly on average, while SNAP low-access shares decreased slightly, and the median change was zero across both rural and urban census tracts.

Individual Contributions

Jiayi Lu: Responsible for data preparation and writing the Exploratory Data Analysis section. Also contributed to the EDA, including summary statistics, plots, and matched census-tract analysis.

Oliver Temel: Responsible for writing the Ethics, Privacy and Security section, report formatting, references, and final editing. Also contributed to the EDA, including summary missing values

Mairangi Campbell: Responsible for write Background and Data section. Also contribute to plots in EDA.

References:

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